WORKERS AND CSVs GOLDEN TRUTHS

WORKERS AND CSVs

WORKER	CSV_FILES	SCHEMA	IN_WORKSCHEDUI
GOLDEN WORKERS			
LateNightScreenRollupWorker	**WRITES:** daily_late_night_screen_usage.csv **READS:** screen_log.csv unlock_log.csv	**daily_late_night_screen_usage.csv:** record_id participant_id date feature_schema_version event late_night	✅ YES
NotificationEngagementWorker	**WRITES:** daily_notification_engagement.csv notification_engagement_heartbeat.csv **READS:** notification_log.csv	**daily_notification_engagement.csv:** record_id participant_id date feature_schema_version event notif_posted notif_engaged notification_engagement_rate notif_latency_avg_s notif_latency_median_s **notification_engagement_heartbeat.csv:** ts message **notification_log.csv:** ts event_type package_name notification_id raw_event raw_reason	✅ YES (WorkScheduler.kt NotificationEngage
UsageEntropyDailyWorker	**WRITES:** daily_usage_entropy.csv **READS:** usage_diag.csv	**daily_usage_entropy.csv:** record_id participant_id date feature_schema_version event usage_entropy **usage_diag.csv:** date mode status	✅ YES (WorkScheduler.kt UsageEntropyDaily
UsageCaptureWorker	**WRITES:** daily_app_usage_minutes.csv usage_capture_log.csv **READS:** (none)	**daily_app_usage_minutes.csv:** record_id participant_id date feature_schema_version event app_min_social app_min_dating	✅ YES (WorkScheduler.kt UsageWorkers.kt re UsageCaptureWorl

WORKER	CSV_FILES	SCHEMA	IN_WORKSCHEDUI
		app_min_productivity app_min_music_audio app_min_image app_min_maps app_min_video app_min_travel_local app_min_shopping app_min_news app_min_game app_min_health app_min_finance app_min_browser app_min_comm app_min_other app_min_total **usage_capture_log.csv:** ts tag	
DistanceSummaryWorker	**WRITES**: daily_distance_log.csv **READS**: location_log.csv	**daily_distance_log.csv:** record_id participant_id date feature_schema_version event distance_m **location_log.csv:** (timestamp in format "YYYY-MM-DD HH:mm:ss") lat lon acc	✅ YES (MainApplication.kt DistanceSummary\ at 3:10 AM)
MovementIntensityDailyWorker	**WRITES**: daily_movement_intensity.csv movement_intensity_heartbeat.csv **READS**: unlock_log.csv	**daily_movement_intensity.csv:** record_id participant_id date feature_schema_version event move_intensity_score **movement_intensity_heartbeat.csv:** ts message **unlock_log.csv:** ts event	✅ YES (WorkScheduler.kt MainApplication.kt MovementIntensity MainApplication.kt daily at 4:45 AM)
SERVICES (NOT WORKERS)			
ForegroundUnlockService	**WRITES:** unlock_log.csv unlock_diag.csv heartbeat.csv screen_log.csv	**unlock_log.csv:** ts event **unlock_diag.csv:** ts tag	N/A (Service)

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	READS: (none)	extra **heartbeat.csv:** ts **screen_log.csv:** ts state	
NotificationLogService	**WRITES:** notification_log.csv **READS:** None	**notification_log.csv:** ts event_type package_name notification_id raw_event raw_reason	❌ NO (registered NotificationListene a Worker)
LocationCaptureService	**WRITES**: location_log.csv	**location_log.csv:** (timestamp in format "YYYY-MM-DD HH:mm:ss") lat lon acc	✅ YES
VALIDATION/DIAGNOSTIC WORKERS			
HealthSnapshotWorker	**WRITES**: daily_health.csv **READS**: unlock_log.csv daily_unlocks.csv redcap_upload_log.csv	**daily_health.csv:** ts unlocks_raw_rows unlocks_rollup_rows last_upload_status	✅ YES
DailySummaryWorker	**WRITES**: daily_summary.csv **READS**: unlock_diag.csv screen_log.csv notification_log.csv daily_app_usage_minutes.csv	**daily_summary.csv:** date total_unlocks battery_drain_rate screen_usage_min notification_count app_usage_min	✅ YES (MainApplication.kt DailySummaryWorl 3:00 AM)
RedcapUploadWorker	**WRITES**: daily_metrics_upload.csv (encrypted) redcap_receipts.csv **READS**: All daily_*.csv files (scans filesDir for files matching pattern)	**daily_metrics_upload.csv (encrypted payload):** record_id participant_id date feature_schema_version daily_unlocks late_night notif_posted notif_engaged notif_engagement_rate notif_latency_avg_s notif_latency_median_s notif_latency_n usage_events_count app_min_social	✅ YES

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		app_min_dating app_min_productivity app_min_music_audio app_min_image app_min_maps app_min_video app_min_travel_local app_min_shopping app_min_news app_min_game app_min_health app_min_finance app_min_other app_min_total app_switch_count usage_entropy distance_m move_intensity_score daily_metrics_complete **redcap_receipts.csv:** ts rows http note	
UsageAccessDiagWorker	**WRITES**: usage_diag.csv **READS**: (none)	**usage_diag.csv:** date mode status	✅ YES
UnlockValidationWorker	**WRITES**: qa_YYYY-MM-DD_unlocks.json (JSON output, not CSV) **READS**: daily_unlocks.csv (deprecated file) unlock_log.csv	**qa_YYYY-MM-DD_unlocks.json (JSON):** feature date exists_daily_unlocks exists_unlock_log pass reasons (array)	✅ YES
RollupValidator	**WRITES**: qa/YYYY-MM-DD/[feature].json (JSON output, not CSV) **READS**: daily_unlocks.csv (deprecated file - via validateUnlocks() method) [any CSV file passed to validate() method]	**qa/YYYY-MM-DD/[feature].json (JSON):** pass (boolean) reasons (array) stats: rows dup_keys bad_cols bad_types bad_range today_rows header	N/A
RedcapDebugReceiver	**WRITES**: redcap_diag.csv **READS**: (none)	**redcap_diag.csv:** ts event	✅ YES (BroadcastReceive by debug actions)
redcap_upload_log.csv	**WRITER**: `RetryQueue.kt` (via `logUpload()` function) **READERS**: `HealthSnapshotWorker.kt` (reads last upload status)	date http_code ok body_snippet	N/A

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usage_diag.csv	**WRITER**: `UsageAccessDiagWorker.kt` **READERS**: None found (yet - may be read by other components)	date mode status	N/A
SUPPORT WORKERS			
LogExportWorker	**WRITES**: log_export.csv export_logs.zip (ZIP file containing all CSV files) **READS**: ALL CSV files in filesDir (bundles them into zip)	**log_export.csv:** date csv_count zip_bytes status **export_logs.zip:** Contains copies of all CSV files found in filesDir	❌ NO A user-initiated act scheduled backgro
UsageEventsDumpWorker	**WRITES**: usage_events.csv **READS**: None (reads from UsageStatsManager system service)	**usage_events.csv:** date time event_type package	✅ YES (MainApplication.kt PeriodicWorkReque appears to be peric (OneTimeWorkReq with tag "${TAG_USAGE_EV for catch-up))
UsageEventsDailyWorker	**WRITES**: None **READS**: None	N/A	✅ YES
ServiceHealthCheckWorker	**WRITES**: service_health_check.csv **READS**: None	**service_health_check.csv:** ts was_running action_taken	✅ YES
NotificationValidationWorker	**WRITES**: None **READS**: daily_notification_engagement.csv daily_notification_latency.csv	N/A	✅ YES
StartServiceWorker	**WRITES**: None **READS**: None	N/A	❌ NO Boot-triggered initi task, not a periodic
RingerChangeReceiver	**WRITES**: ringer_log.csv **READS**: None	**ringer_log.csv:** ts mode	N/A
UnlockMigrations	**WRITES**: migrations.log daily_unlocks.csv **READS**:	**migrations.log:** ts feature from to	N/A

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	unlock_rollup.csv daily_unlocks_count.csv daily_unlocks_v0.csv	rows **daily_unlocks.csv:** (consolidated from legacy files)	
ServiceStarter	N/A	N/A	✅ YES